Understanding CUPED

towardsdatascience.com/understanding-cuped-a822523641af

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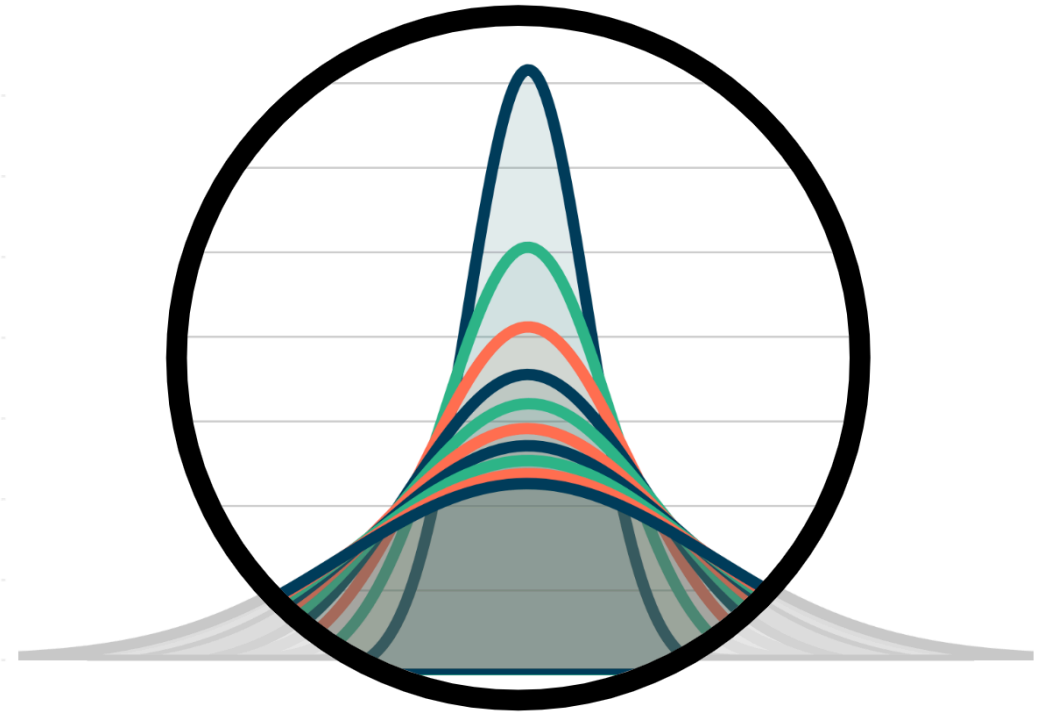
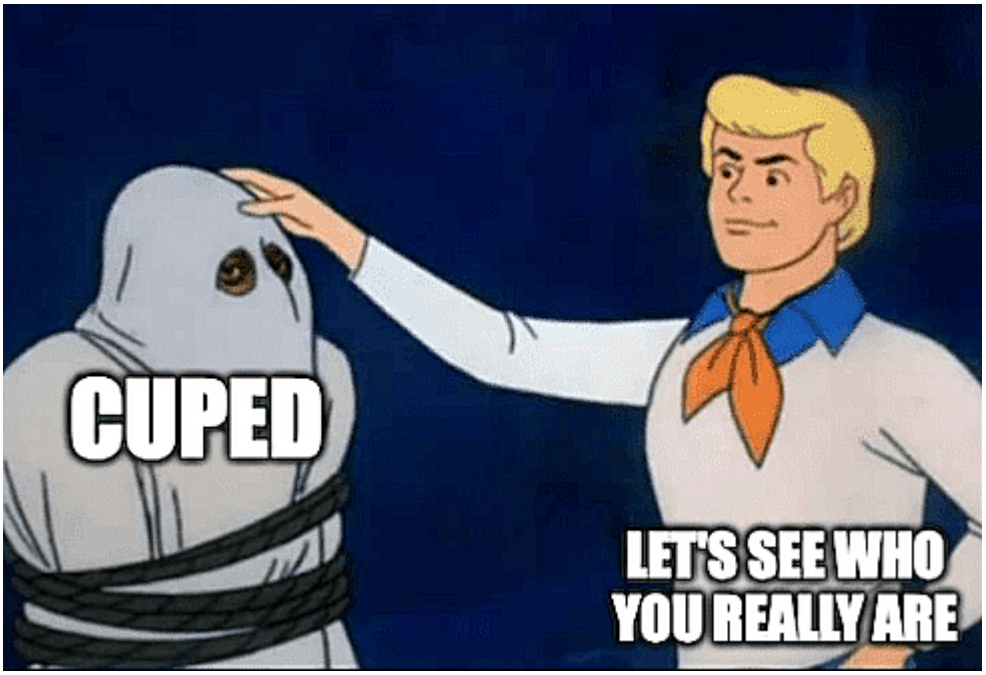
CAUSAL DATA SCIENCE



Cover, image by Author

During my Ph.D., I spent a lot of time learning and applying causal inference methods to experimental and observational data. However, I was completely clueless when I first heard of **CUPED** (Controlled-Experiment using Pre-Experiment Data), a technique to increase the power of randomized controlled trials in A/B tests.

What really amazed me was the popularity of the algorithm in the industry. CUPED was first introduced by Microsoft researchers Deng, Xu, Kohavi, Walker (2013) and has been **widely used in companies** such as Netflix, Booking, Meta, Uber, Airbnb, Linkedin, TripAdvisor, DoorDash, Faire, and many others. While digging deeper into it, I noticed a similarity with some causal inference methods I was familiar with, such as Difference-in-Differences or regression with control variables. I was curious and decided to dig deeper.



Who are you CUPED? Image by Author

TLDR; CUPED is essentially a [residualized outcome regression]. It is generally not equivalent to neither diff-in-diffs, nor to regression with control variables, except for special cases.

Example

Let's assume we are a firm that is testing an and we are interested in understanding whether it increases revenue or not. We randomly split a set of users into a treatment and control group and we show the ad campaign to the treatment group. Differently from the standard A/B test setting, assume we observe users also before the test.

We can now generate the simulated data, using the data generating process `dgp_cuped()` from `[src.dgp](https://github.com/matteocourthoud/Blog-Posts/blob/main/notebooks/src/dgp.py)`. I also import some plotting functions and libraries from `[src.utils](https://github.com/matteocourthoud/Blog-Posts/blob/main/notebooks/src/utils.py)`.

```
from src.utils import *
from src.dgp import dgp_cuped

df = dgp_cuped().generate_data()
df.head()
```

	i	ad_campaign	revenue0	revenue1
0	1	0	5.315635	8.359304
1	2	1	2.977799	7.751485
2	3	0	4.693796	9.025253
3	4	0	5.827975	8.540667
4	5	0	5.230095	8.910165

Data snapshot, image by Author

We have information on 1000 individuals indexed by `i` for whom we observe the revenue generated pre and post-treatment, `revenue0` and `revenue1` respectively, and whether they have been exposed to the `ad_campaign`.

Difference in Means

In randomized experiments or A/B tests, **randomization** allows us to estimate the average treatment effect using a simple difference in means. We can just compare the average outcome post-treatment Y_1 (`revenue1`) across control and treated units and randomization guarantees that this difference is due to the treatment alone, in expectation.

Simple difference estimator, image by Author

Where the bar indicates the average over individuals and the subscript *d* indicates the treatment status. In our case, we compute the average revenue post ad campaign in the treatment group, minus the average revenue post ad campaign in the control group.

```
np.mean(df.loc[df.ad_campaign==True, 'revenue1']) -
np.mean(df.loc[df.ad_campaign==False, 'revenue1'])
```

```
1.7914301325347406
```

The estimated treatment effect is 1.79, very close to the **true value** of 2. We can obtain the same estimate by regressing the post-treatment outcome `revenue1` on the treatment indicator *D* for `ad_campaign`.

Regression equation, image by Author

Where β is the coefficient of interest.

```
smf.ols('revenue1 ~ ad_campaign', data=df).fit().summary().tables[1]
```

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	8.2995	0.211	39.398	0.000	7.881	8.718
ad_campaign	1.7914	0.301	5.953	0.000	1.194	2.389

Regression results, image by Author

This estimator is **unbiased**, which means it delivers the correct estimate, on average. However, it can still be improved: we could **decrease its variance**. Decreasing the variance of an estimator is extremely important since it allows us to

- **detect smaller effects**
- detect the same effect, but with a **smaller sample size**

In general, an estimator with a smaller variance allows us to run tests with a higher **power**, i.e. ability to detect smaller effects.

Can we improve the power of our AB test? Yes, with CUPED (among other methods).

CUPED

The **idea** of CUPED is the following. Suppose you are running an AB test and *Y* is the outcome of interest (`revenue` in our example) and the binary variable *D* indicates whether a single individual has been treated or not (`ad_campaign` in our example).

Suppose you have access to another random variable *X* which is **not affected** by the treatment and has known expectation $\mathbb{E}[X]$. Then define

Transformed outcome, image by Author

where θ is a scalar. This estimator is an **unbiased** estimator for $\mathbb{E}[Y]$ since in expectation the two last terms cancel out. However, the variance of $\hat{Y}_1^{cuped}$ is

$$\begin{aligned}\mathrm{Var}\left(\hat{Y}_1^{cuped}\right) &= \mathrm{Var}\left(\bar{Y}_1 - \theta\bar{X} + \theta\mathbb{E}[X]\right) = \\ &= \mathrm{Var}\left(Y_1 - \theta X\right)/n = \\ &= \Big(\mathrm{Var}(Y_1) + \theta^2\mathrm{Var}(X) - 2\theta\mathrm{Cov}(X,Y)\Big)/n\end{aligned}$$

Transformed outcome variance, image by Author

Note that the variance of $\hat{Y}_1^{cuped}$ is minimized for

$$\theta^* = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$$

Optimal theta, image by Author

Which is the **OLS** estimator of a linear regression of *Y* on *X*. Substituting θ^* into the formula of the variance of $\hat{Y}_1^{cuped}$ we obtain

$$\mathrm{Var}\left(\hat{Y}_1^{cuped}\right) = \mathrm{Var}(\bar{Y})(1-\rho^2)$$

Transformed outcome variance, image by Author

where ρ is the **correlation** between Y and X. Therefore, the higher the correlation between Y and X, the higher the variance reduction of CUPED.

We can then **estimate the average treatment effect** as the average difference in the transformed outcome between the control and treatment groups.

$$\begin{aligned}\widehat{ATE}^{cuped} &= \hat{Y}_1^{cuped}(D=1) - \hat{Y}_1^{cuped}(D=0) = \\ &= \left(\bar{Y}_1 - \theta\bar{X} + \theta\mathbb{E}[X] \,\middle|\, D=1\right) - \left(\bar{Y}_1 - \theta\bar{X} + \theta\mathbb{E}[X] \,\middle|\, D=0\right) = \\ &= \left(\bar{Y}_1 - \theta\bar{X} \,\middle|\, D=1\right) - \left(\bar{Y}_1 - \theta\bar{X} \,\middle|\, D=0\right)\end{aligned}$$

CUPED estimator, image by Author

Note that $\mathbb{E}[X]$ cancels out when taking the difference. Therefore, it is sufficient to compute

Equivalent formulation of the transformed outcome, image by Author

This is not an unbiased estimator of $\mathbb{E}[Y]$ but still delivers an unbiased estimator of the average treatment effect.

Optimal X

What is the **optimal choice** for the control variable X?

We know that X should have the following **properties**:

- not affected by the treatment
- as correlated with Y_1 as possible

The authors of the paper suggest using the **pre-treatment outcome** Y_0 since it gives the most variance reduction in practice.

To **summarize**, we can compute the CUPED estimate of the average treatment effect as follows:

1. Regress Y_1 on Y_0 and estimate $\hat{\theta}$
2. Compute $\hat{Y}_1^{cuped} = Y_1 - \hat{\theta}\, Y_0$
3. Compute the difference of $\hat{Y}_1^{cuped}$ between treatment and control group

Equivalently, we can compute $\hat{Y}_1^{cuped}$ at the individual level and then regress it on the treatment dummy variable D.

Back To The Data

Let's compute the CUPED estimate for the treatment effect, one step at a time. First, let's estimate θ.

```
theta = smf.ols('revenue1 ~ revenue0', data=df).fit().params[1]
```

Now we can compute the transformed outcome $\hat{Y}_1^{cuped}$.

```
df['revenue1_cuped'] = df['revenue1'] - theta * (df['revenue0'] -
np.mean(df['revenue0']))
```

Lastly, we estimate the treatment effect as a difference in means, with the transformed outcome $\hat{Y}_1^{cuped}$.

```
smf.ols('revenue1_cuped ~ ad_campaign', data=df).fit().summary().tables[1]
```

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	8.2259	0.143	57.677	0.000	7.943	8.509
ad_campaign	1.9415	0.204	9.529	0.000	1.537	2.346

Regression results, image by Author

The standard error is 33% smaller (0.2 vs 0.3)!

Equivalent Formulation

An alternative but algebraically **equivalent** way of obtaining the CUPED estimate is the following

1. Regress Y_1 on Y_0 and compute the residuals $\tilde{Y}_1$
2. Compute $\hat{Y}_1^{cuped} = \tilde{Y}_1 + \bar{Y}_1$
3. Compute the difference of $\hat{Y}_1^{cuped}$ between the treatment and control group

Step (3) is the same as before but (1) and (2) are different. This procedure is called **partialling out** and the algebraic equivalence is guaranteed by the [Frisch-Waugh-Lowell Theorem].

Let's check if we indeed obtain the same result.

```
df['revenue1_tilde'] = smf.ols('revenue1 ~ revenue0', data=df).fit().resid +
np.mean(df['revenue1'])
```

```
smf.ols('revenue1_tilde ~ ad_campaign', data=df).fit().summary().tables[1]
```

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	8.2259	0.143	57.677	0.000	7.943	8.509
ad_campaign	1.9415	0.204	9.529	0.000	1.537	2.346

Regression results, image by Author

Yes! The regression table is exactly identical.

CUPED vs Other

CUPED seems to be a very powerful procedure but it is remindful of at least a couple of other methods.

1. **Autoregression** or regression with control variables
2. **Difference-in-Differences**

Are these methods the same or is there a difference? Let's check.

Autoregression

The **first question** that came to my mind when I first saw CUPED was "*is CUPED just the simple difference with an additional control variable?*". Or equivalently, is CUPED equivalent to estimating the following regression via least squares (where γ is the parameter of interest)?

Estimating equation, image by Author

Let's have a look.

```
smf.ols('revenue1 ~ revenue0 + ad_campaign', data=df).fit().summary().tables[1]
```

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	1.9939	0.603	3.304	0.001	0.796	3.192
revenue0	1.2249	0.114	10.755	0.000	0.999	1.451
ad_campaign	1.9519	0.205	9.529	0.000	1.545	2.358

Regression results, image by Author

The estimated coefficient is very similar to the one we obtained with CUPED and also the standard error is very close. However, they are **not exactly the same**.

If you are familiar with the Frisch-Waugh-Lowell Theorem, you might wonder why the two procedures are **not equivalent**. The reason is that with CUPED we are partialling out only Y, while the FWL theorem holds when we are partialling out either X or both X and Y.

Diff-in-Diffs

The **second question** that came to my mind was "i*s CUPED just difference-in-differences?*". Difference-in-Differences (or diff-in-diffs, or DiD) is an estimator that computes the treatment effect as a **double-difference** instead of a single one: pre-post and treatment-control instead of just treatment-control.

Difference-in-differences estimator, image by Author

This method was initially introduced by [John Snow] (https://en.wikipedia.org/wiki/John*Snow*(disambiguation)) (1854) (no, not that John Snow) to estimate the causes of a Cholera epidemic in London. The main advantage of diff-in-diff is that it allows to estimate the average treatment effect when randomization is not perfect and the treatment and control group are not comparable. The **key assumption** is that the difference between the treatment and control groups is constant over time. By taking a double difference, we cancel it out.

Let's check how diff-in-diffs works empirically. The most common way to compute the diff-in-diffs estimator is to first reshape the data in a **long format** or **panel format** (one observation is an individual *i* at time period *t*) and then regress the outcome *Y* on the full interaction between the post-treatment dummy $\mathbb{I}(t=1)$ and the treatment dummy D.

Estimating equation, image by Author

The estimator of the average treatment effect is the coefficient of the interaction coefficient, δ.

```
df_long = pd.wide_to_long(df, stubnames='revenue', i='i', j='t').reset_index()
df_long.head()
```

	i	t	ad_campaign	revenue1_tilde	revenue1_cuped	revenue
0	1	0	0	8.093744	8.093744	5.315635
1	2	0	1	10.164644	10.164644	2.977799
2	3	0	0	9.472203	9.472203	4.693796
3	4	0	0	7.688063	7.688063	5.827975
4	5	0	0	8.742618	8.742618	5.230095

Long dataset snapshot, image by Author

The long dataset is now indexed by individuals `i` and time `t`. We are now ready to run the diff-in-diffs regression.

```
smf.ols('revenue ~ t * ad_campaign', data=df_long).fit().summary().tables[1]
```

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	5.1481	0.174	29.608	0.000	4.805	5.491
t	3.1514	0.246	12.816	0.000	2.666	3.636
ad_campaign	-0.1310	0.248	-0.527	0.599	-0.621	0.359
t:ad_campaign	1.9224	0.351	5.473	0.000	1.230	2.615

Regression results, image by Author

The estimated coefficient is close to the true value, 2, but the standard errors are bigger than the ones obtained with all other methods (0.35 >> 0.2). What did we miss? We didn't **cluster the standard errors**!

I won't go into detail here on what standard error clustering means, but the intuition is the following. The `statsmodels` package by default computes the standard errors assuming that outcomes are **independent** across observations. This assumption is unlikely to be true in this setting where we observe individuals over time and we are trying to exploit this information. Clustering allows for **correlation** of the outcome variable within clusters. In our case, it makes sense (even without knowing the data generating process) to cluster the standard errors at the individual level, allowing the outcome to be correlated over time for an individual *i*.

```
smf.ols('revenue ~ t * ad_campaign', data=df_long)
    .fit(cov_type='cluster', cov_kwds={'groups': df_long['i']})
    .summary().tables[1]
```

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	5.1481	0.139	37.056	0.000	4.876	5.420
t	3.1514	0.128	24.707	0.000	2.901	3.401
ad_campaign	-0.1310	0.181	-0.724	0.469	-0.486	0.224
t:ad_campaign	1.9224	0.209	9.208	0.000	1.513	2.332

Regression results, image by Author

Clustering standard errors at the individual level we obtain standard errors that are comparable to the previous estimates (~0.2).

Note that diff-in-diffs is **equivalent to CUPED** when we assume the CUPED coefficient $\theta=1$.

```
df['revenue1_cuped2'] = df['revenue1'] - 1 * (df['revenue0'] - np.mean(df['revenue0']))
smf.ols('revenue1_cuped2 ~ ad_campaign', data=df).fit().summary().tables[1]
```

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	8.2353	0.145	56.756	0.000	7.947	8.523
ad_campaign	1.9224	0.207	9.274	0.000	1.511	2.334

Regression results, image by Author

Indeed, we obtain the same exact coefficient and almost identical standard errors!

Comparison

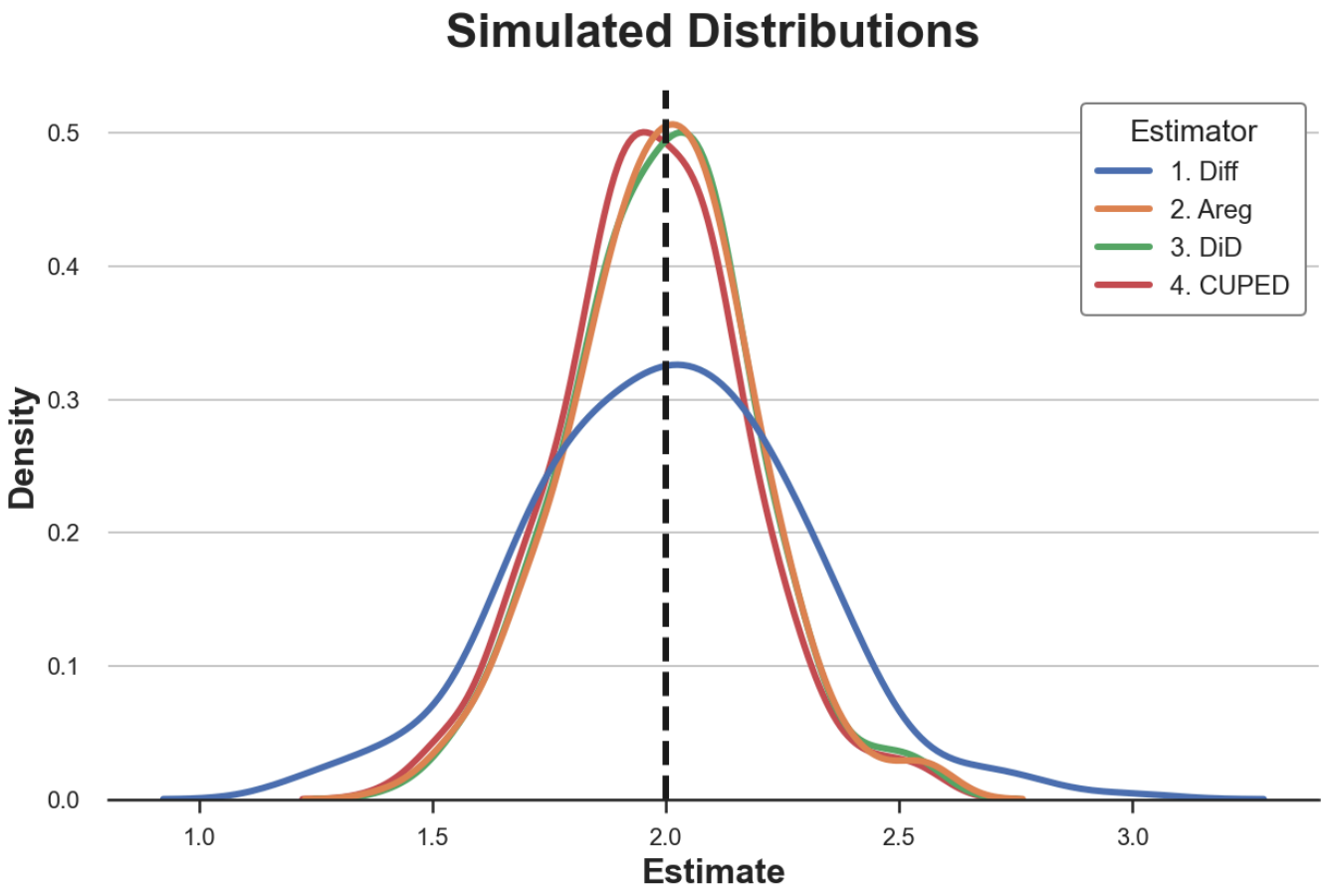
Which method is better? From what we have seen so far, all methods seem to deliver an accurate estimate, but the simple difference has a larger standard deviation.

Let's now compare all the methods we have seen so far via **simulation**. We simulate the data generating process `dgp_cuped()` 1000 times and we save the estimated coefficient of the following methods:

1. Simple difference
2. Autoregression
3. Diff-in-diffs
4. CUPED

Let's plot the distribution of the estimated parameters.

```
sns.kdeplot(data=results, x="Estimate", hue="Estimator");
plt.axvline(x=2, c='k', ls='--');
plt.title('Simulated Distributions');
```



Simulated distribution of estimates, image by Author

We can also tabulate the simulated mean and standard deviation of each estimator.

```
results.groupby('Estimator').agg(mean=("Estimate", "mean"), std=("Estimate", "std"))
```

Estimator	mean	std
1. Diff	1.999626	0.291497
2. Areg	1.993343	0.198337
3. DiD	1.993494	0.197638
4. CUPED	1.971853	0.198145

Summary table, image by Author

All estimators seem **unbiased**: the average values are all close to the true value of 2. Moreover, all estimators have a very similar standard deviation, apart from the single-difference estimator!

Always Identical?

Are the estimators always identical, or is there some difference among them?

We could check many different departures from the original data generating process. For simplicity, I will consider only one here: **imperfect randomization**. Other tweaks to the data generating process that I considered are:

- pre-treatment missing values
- additional covariates / control variables
- multiple pre-treatment periods
- heterogeneous treatment effects

and combinations of them. However, I found imperfect randomization to be the most informative example.

Suppose now that **randomization was not perfect** and two groups are not identical. In particular, if the data generating process is

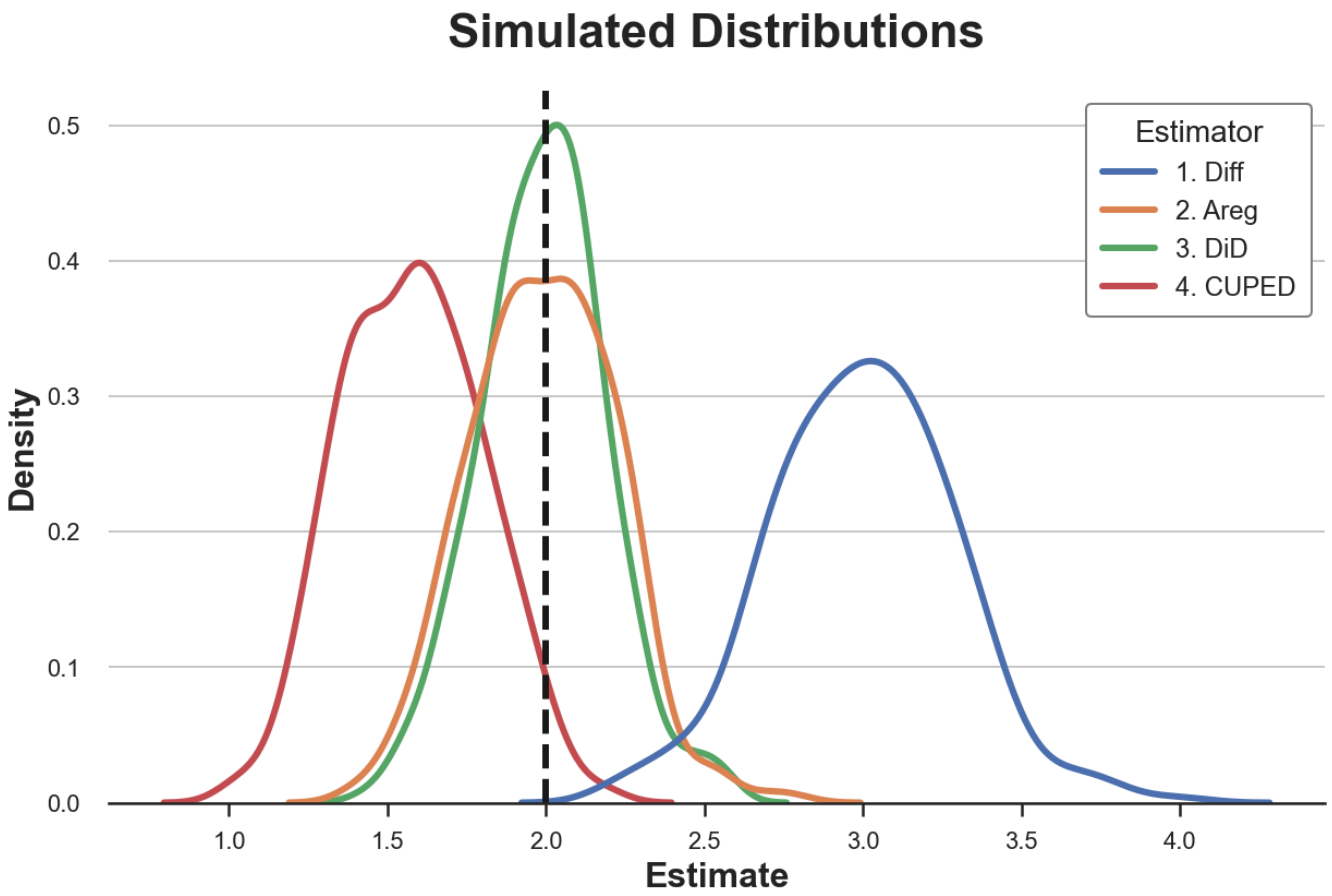
Data generating process, image by Author

assume that $\beta \neq 0$. Also, note that we have **persistent individual-level heterogeneity** because the unobservable u_i does not change over time (is not indexed by t).

```
results_beta1 = simulate(dgp=dgp_cuped(beta=1))
```

Let's plot the distribution of the estimated parameters.

```
sns.kdeplot(data=results_beta1, x="Estimate", hue="Estimator");
plt.axvline(x=2, c='k', ls='--');
plt.title('Simulated Distributions');
```



Simulated distribution of estimates, image by Author

```
results_beta1.groupby('Estimator').agg(mean=("Estimate", "mean"), std=("Estimate",
"std"))
```

Estimator	mean	std
1. Diff	2.999626	0.291497
2. Areg	1.991508	0.227065
3. DiD	1.993494	0.197638
4. CUPED	1.577712	0.221448

Summary table, image by Author

With imperfect treatment assignment, both difference-in-differences and autoregression are **unbiased** for the true treatment effect, however, diff-in-diffs is **more efficient**. Both CUPED and simple difference are **biased** instead. Why?

Diff-in-diffs explicitly controls for **systematic differences** between the treatment and control group that are **constant over time**. This is exactly what this estimator was built for. Autoregression performs some sort of matching on the additional covariate, Y_0, effectively controlling for these systematic differences, but less efficiently (if you want to know more, I wrote related posts on control variables [here](https://towardsdatascience.com/58b63d25d2ef)

and here). CUPED controls for persistent heterogeneity at the individual level, but not at the treatment assignment level. Lastly, the simple difference estimator does not control for anything.

Conclusion

In this post, I have analyzed an estimator of average treatment effects in AB testing, very popular in the industry: CUPED. The key idea is that, by exploiting pre-treatment data, CUPED can **achieve a lower variance** by controlling for individual-level variation that is persistent over time. We have also seen that CUPED is closely related but not equivalent to autoregression and difference-in-differences. The differences among the methods clearly emerge when we have imperfect randomization.

An interesting avenue of future research is what happens when we have **a lot of pre-treatment information**, either in terms of time periods or observable characteristics. Scientists from Meta, Guo, Coey, Konutgan, Li, Schoener, Goldman (2021), have analyzed this problem in a very recent paper that exploits machine learning techniques to efficiently use this extra information. This approach is closely related to the Double/Debiased Machine Learning literature. If you are interested, I wrote two articles on the topic (part 1 and part 2) and I might write more in the future.

References

[1] A. Deng, Y. Xu, R. Kohavi, T. Walker, Improving the Sensitivity of Online Controlled Experiments by Utilizing Pre-Experiment Data (2013), *WSDM*.

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[3] Y. Guo, D. Coey, M. Konutgan, W. Li, C. Schoener, M. Goldman, Machine Learning for Variance Reduction in Online Experiments (2021), *NeurIPS*.

[4] V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, J. Robins, Double/debiased machine learning for treatment and structural parameters (2018), *The Econometrics Journal*.

[5] M. Bertrand, E. Duflo, S. Mullainathan, How Much Should We Trust Differences-In-Differences Estimates? (2012), *The Quarterly Journal of Economics*.

Related Articles

- Double Debiased Machine Learning (part 1)
- Double Debiased Machine Learning (part 2)
- Understanding The Frisch-Waugh-Lovell Theorem
- Understanding Contamination Bias
- DAGs and Control Variables

Code

You can find the original Jupyter Notebook here:

> **Blog-Posts/cuped.ipynb at main · matteocourthoud/Blog-Posts**

Thank you for reading!

*I really appreciate it! If you liked the post and would like to see more, consider f**ollowing me.** I post once a week on topics related to causal inference and data analysis. I try to keep my posts simple but precise, always providing code, examples, and simulations.*

*Also, a small **disclaimer**: I write to learn so mistakes are the norm, even though I try my best. Please, when you spot them, let me know. I also appreciate suggestions on new topics!*